EXP NO: 5	Comparative Analysis Of Vgg, Resnet, And Googlenet

AIM:

To compare the performance of the pre-trained VGG16, ResNet18 and GoogLeNet architectures on CIFAR-10 by fine-tuning each of them on a small subset of the dataset.

ALGORITHM:

1. Load CIFAR-10, resize the images to 224 x 224, normalize them, use the first 500 images and split them 80% / 20% into training and validation sets (batch size 32).
2. Create each model with ImageNet pre-trained weights (VGG16, ResNet18, GoogLeNet) and replace its final layer with a 10-class linear layer.
3. Train each model for 2 epochs using the Adam optimizer (learning rate 0.0001) and the cross-entropy loss.
4. After every epoch compute the training accuracy and the validation accuracy.
5. Print the final training and validation accuracy of the three models and plot their validation accuracy.
6. Select the best model based on the final validation accuracy.

CODE:

```
import torch
import torch.nn as nn
import torch.optim as optim

from torchvision import datasets
from torchvision import models
from torchvision import transforms

from torch.utils.data import DataLoader
from torch.utils.data import random_split
from torch.utils.data import Subset

import matplotlib.pyplot as plt

print("PyTorch version:", torch.__version__)

device = torch.device(
    "cuda" if torch.cuda.is_available() else "cpu"
)

print("Using device:", device)

class_names = [
    "airplane",
    "automobile",
    "bird",
    "cat",
    "deer",
    "dog",
    "frog",
    "horse",
    "ship",
    "truck"
]

transform = transforms.Compose([
    transforms.Resize((224, 224)),
```

```
    transforms.ToTensor(),
    transforms.Normalize(
        (0.5, 0.5, 0.5),
        (0.5, 0.5, 0.5)
    )
])

dataset = datasets.CIFAR10(
    root="./data",
    train=True,
    download=True,
    transform=transform
)

# Use 500 images as in the lab manual
dataset = Subset(
    dataset,
    range(500)
)

train_size = int(
    0.8 * len(dataset)
)
val_size = len(dataset) - train_size

train_dataset, val_dataset = random_split(
    dataset,
    [train_size, val_size]
)

train_loader = DataLoader(
    train_dataset,
    batch_size=32,
    shuffle=True
)

val_loader = DataLoader(
    val_dataset,
    batch_size=32,
    shuffle=False
)

def create_model(name):

    if name == "vgg":
        model = models.vgg16(
            weights=models.VGG16_Weights.DEFAULT
        )
        model.classifier[6] = nn.Linear(
            4096,
            10
        )

    elif name == "resnet":
        model = models.resnet18(
            weights=models.ResNet18_Weights.DEFAULT
        )
        model.fc = nn.Linear(
            model.fc.in_features,
            10
        )

    elif name == "googlenet":
```

```
        model = models.googlenet(
            weights=models.GoogLeNet_Weights.DEFAULT,
            aux_logits=True
        )
        model.fc = nn.Linear(
            model.fc.in_features,
            10
        )

    else:
        raise ValueError(
            "Unknown model name"
        )

    return model

def train_and_evaluate(
    model,
    name,
    epochs=2
):

    model = model.to(device)

    criterion = nn.CrossEntropyLoss()

    optimizer = optim.Adam(
        model.parameters(),
        lr=0.0001
    )

    train_accuracies = []
    val_accuracies = []

    print(
        f"\n===== {name} ====="
    )

    for epoch in range(epochs):

        model.train()

        correct = 0
        total = 0

        for inputs, labels in train_loader:

            inputs = inputs.to(device)
            labels = labels.to(device)

            optimizer.zero_grad()

            outputs = model(inputs)

            if isinstance(outputs, tuple):
                outputs = outputs[0]

            loss = criterion(
                outputs,
                labels
            )

            loss.backward()
```

```
            optimizer.step()

            _, predicted = torch.max(
                outputs,
                1
            )

            total += labels.size(0)

            correct += (
                predicted == labels
            ).sum().item()

        train_accuracy = (
            100 * correct / total
        )

        train_accuracies.append(
            train_accuracy
        )

        model.eval()

        correct = 0
        total = 0

        with torch.no_grad():

            for inputs, labels in val_loader:

                inputs = inputs.to(device)
                labels = labels.to(device)

                outputs = model(inputs)

                if isinstance(outputs, tuple):
                    outputs = outputs[0]

                _, predicted = torch.max(
                    outputs,
                    1
                )

                total += labels.size(0)

                correct += (
                    predicted == labels
                ).sum().item()

        val_accuracy = (
            100 * correct / total
        )

        val_accuracies.append(
            val_accuracy
        )

        print(
            f"Epoch {epoch + 1}/{epochs} "
            f"- Train Accuracy: "
            f"{train_accuracy:.2f}% "
            f"- Validation Accuracy: "
            f"{val_accuracy:.2f}%"
```

```
        )

    return (
        model,
        train_accuracies,
        val_accuracies
    )

results = {}
trained_models = {}

for model_name in [
    "vgg",
    "resnet",
    "googlenet"
]:

    model = create_model(model_name)

    trained_model, train_acc, val_acc = train_and_evaluate(
        model,
        model_name.upper(),
        epochs=2
    )

    results[model_name] = (
        train_acc,
        val_acc
    )

    trained_models[model_name] = trained_model

print("\n===== FINAL RESULTS =====")

for name, values in results.items():

    train_acc = values[0]
    val_acc = values[1]

    print(
        f"{name.upper():10s} "
        "Final Train Accuracy: "
        f"{train_acc[-1]:.2f}% "
        "| Final Validation Accuracy: "
        f"{val_acc[-1]:.2f}%"
    )

plt.figure(figsize=(18, 6))

for name, values in results.items():

    train_acc = values[0]
    val_acc = values[1]

    plt.plot(
        range(1, len(val_acc) + 1),
        val_acc,
        marker="o",
        label=f"{name.upper()} Validation Accuracy"
    )

plt.xlabel("Epoch")
plt.ylabel("Accuracy (%)")
```

```
plt.title(
    "VGG vs ResNet vs GoogLeNet"
)
plt.legend()
plt.grid(True)
plt.show()

best_model = max(
    results,
    key=lambda name: results[name][1][-1]
)

print(
    f"\nBest model based on validation accuracy: "
    f"{best_model.upper()}"
)

print(
    f"Validation Accuracy: "
    f"{results[best_model][1][-1]:.2f}%"
)

print("\nExperiment 5 completed successfully.")
```

OUTPUT:

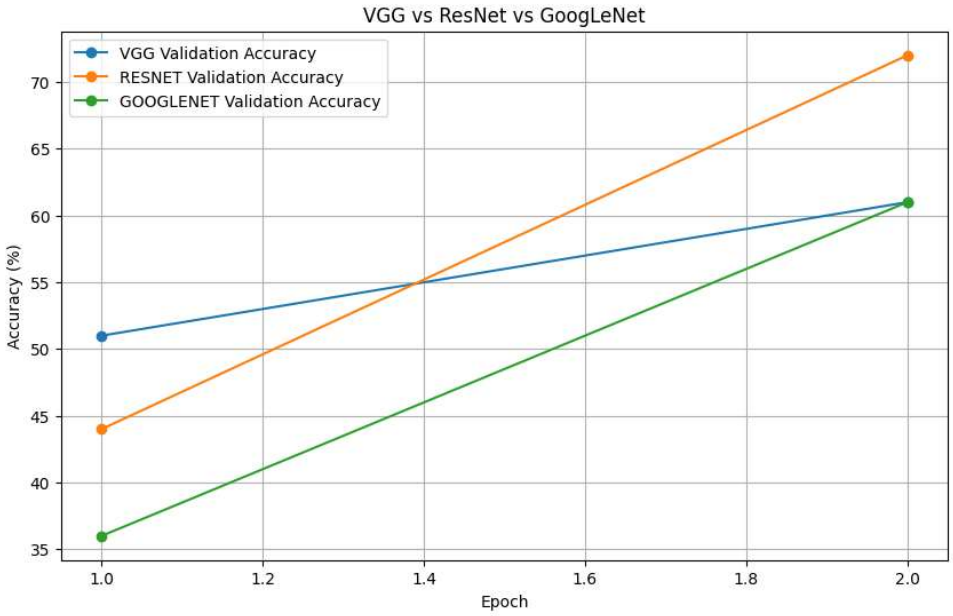
```
PyTorch version: 2.11.0+cpu
Using device: cpu
100%|██████████| 170M/170M [06:07<00:00, 464kB/s]
Downloading: "https://download.pytorch.org/models/vgg16-397923af.pth" to /root/.cache/torch/hub/checkpoints/vgg16-397923a
100%|██████████| 528M/528M [00:07<00:00, 74.7MB/s]

===== VGG =====
Epoch 1/2 - Train Accuracy: 24.50% - Validation Accuracy: 51.00%
Epoch 2/2 - Train Accuracy: 61.25% - Validation Accuracy: 61.00%
Downloading: "https://download.pytorch.org/models/resnet18-f37072fd.pth" to /root/.cache/torch/hub/checkpoints/resnet18-f
100%|██████████| 44.7M/44.7M [00:00<00:00, 163MB/s]

===== RESNET =====
Epoch 1/2 - Train Accuracy: 26.50% - Validation Accuracy: 44.00%
Epoch 2/2 - Train Accuracy: 93.25% - Validation Accuracy: 72.00%
Downloading: "https://download.pytorch.org/models/googlenet-1378be20.pth" to /root/.cache/torch/hub/checkpoints/googlenet
100%|██████████| 49.7M/49.7M [00:00<00:00, 108MB/s]
/usr/local/lib/python3.13/dist-packages/torchvision/models/googlenet.py:341: UserWarning: auxiliary heads in the pretrain
  warnings.warn(

===== GOOGLENET =====
Epoch 1/2 - Train Accuracy: 18.25% - Validation Accuracy: 36.00%
Epoch 2/2 - Train Accuracy: 74.75% - Validation Accuracy: 61.00%

===== FINAL RESULTS =====
VGG        Final Train Accuracy: 61.25% | Final Validation Accuracy: 61.00%
RESNET     Final Train Accuracy: 93.25% | Final Validation Accuracy: 72.00%
```



RESULT:

VGG16, ResNet18 and GoogLeNet were fine-tuned on a 500-image subset of CIFAR-10 for 2 epochs. The final validation accuracy was 72.00% for ResNet18, 61.00% for VGG16 and 61.00% for GoogLeNet, so ResNet18 performed best in this experiment.